

RESEARCH ARTICLE

Spatiotemporal analysis of emerging water bodies in the Indo-Gangetic Plains to Western India corridor

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ABSTRACT

The rising number and severity of extreme heat events in tandem with the accelerating climate change have become important hazards for urban areas, worsening thermal stress, energy consumption, and environmental degradation. Urban waterscapes, being one of the integral components of blue-green infrastructure, play vital roles in local microclimate moderation and preventing exposure to heat. Two cities having varying Köppen climate zones, namely Lucknow (Cwa: Dry Winter Humid Subtropical) and Ahmedabad (Bsh: Hot Steppe), were selected. The help of Landsat 8 satellite imagery and GIS-based thermal analysis in summer seasons was taken, and land surface temperature (LST) profiles were prepared for selected lakes. Scaling methods and radius of saturation (R_Sat) concept were used to standardize water body dimensions for purpose of comparative analysis of cooling impact in a climate-based manner. Results show Lucknow water bodies reached 100 % thermal efficiency with mean 120 m influence radii, while Ahmedabad water bodies reached only 67 % efficiency by 1.49× scaling in order to realize Lucknow's cooling capability. These findings highlight the critical importance of climate-sensitive landscape design in strategies for reducing urban heat and confirm adaptive planning techniques for climate-responsive sustainable cities in different climatic contexts.

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INTRODUCTION

Climate change is intensifying conditions of extreme heat all over the world, and urban areas are becoming thermal stress hotspots through rapid urbanization and habitat loss. Urban heat islands (UHI) further aggravate this by trapping heat in densely populated urban cities, increasing energy use, health risks, and stress to ecosystems. Urban aquatic bodies such as lakes and ponds play significant roles in moderating microclimates through evaporative cooling and thermal buffering. However, their performance depends very much on climatic factors and therefore need climate-sensitive design and management strategies [1].

Past research has established that water bodies are able to decrease land surface temperature (LST) in urban areas, with the cooling effects ranging from a few hundred meters to several kilometers based on size, morphology, and the extent of surrounding land use. Research in areas like Pune, Riyadh, and Hangzhou has been helpful in understanding the functionality of water bodies as UHI mitigation features. But

most studies currently in use are site-specific case studies that fail to factor in climatic variability. Consequently, comparative studies across the different climatic zones are lacking in order to discern how the same landscape feature responds in different thermal and humidity conditions.

Following this gap in knowledge, the present study formulates a climate-sensitivity scaling framework that standardizes water body sizes and contrasts their thermal effect across climatic zones. Based on the radius of saturation (R_Sat) concept, the study quantifies water bodies' cooling capacity and establishes performance differentials between conflicting climates. The combined approach combines remote sensing, Landsat 8 satellite images, and geospatial thermal mapping to provide a comparable baseline for water bodies in different climatic conditions.

The main objective of this study is to evaluate the climatic context effect on the urban water bodies' thermal efficiency. Lucknow (Dry Winter Humid Subtropical—Cwa) and Ahmedabad (Hot Steppe—Bsh) are the study locations, and

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the research aims to (i) analyze the spatial cooling footprint of selected water bodies, (ii) standardize their effect by scaling procedures, and (iii) find climate-sensitive strategies for maximizing urban heat relief. By explicitly linking climatic sensitivity and landscape performance, this research offers a new evidence base for designing resilient, climate-adaptive urban blue infrastructure.

LITERATURE REVIEW

Climatic Classifications

Climate classification has significant roles in determining the many interplays between climate and ecosystems, water resources, and built environments. Significant urban water bodies have been shown to act as mitigators for Urban Heat Island (UHI) effect in the city of Pune, India, reducing surface land temperatures up to roughly 350 meters from the body's perimeter [2]. For example, after COVID-19, cities like Lucknow and Ahmedabad saw an increase in heat stress and torrid days, the need for climate adaptive urban planning is there [3]. Unlike lakes, aquatic ecosystems have also been disrupted by climate change, as its thermal dynamics have been altered. Floating photovoltaics (FPVs) are modeled to reduce thermal stratification and, therefore, could play a role in climate mitigation [4, 5].

Increasingly, climatic classifications are ecologically informed, and are integrating patterns of biodiversity, and dynamics of ecosystem. For example, [6] used phylogenetic β diversity to identify six different wild bee ecoregions (WBEs) of Chile as a baseline for conservation, with the aims of establishing an early planning document in Chile. Studies among desert biocrust [7] and seagrass ecosystem [8] point out the impacts of 'keystone species' on climate sensitive ecological functions.

Quantitative tools, such as the Aridity Index (AI) are climatic indices that can be used to assess water stress at the regional scales, as seen in drought assessment in Turkey [9] and vegetation – climate interaction in the so called Mediterranean zones [10]. However, some regions may be better indicated by an alternative index, such as water deficit (WD).

Of the climate classification systems, however, this is Köppen's scheme, which is the most widely applied because vegetation patterns are integrated into it [11]. However, for other application Thornthwaite [12] Alisov and Holdridge are still of value. Such as climate responsive tools such as the CBE Clima Tool [13] or dynamic energy models [14] they help create the resilient buildings as per the ECBC and the CPWD guidelines [15].

Landscape Indices and Methods

Urban water bodies are key to alleviate the Urban Heat Island (UHI) effect and for thermal regulation in urban areas. From climal zones and seasons, many studies have confirmed their cooling efficacy. The Qiantang River showed a maximum summer cooling effect of 2 °C in Hangzhou, China, and extended up to more than 2500 meters from the cool region [16]. In Pune, India, LST reduction were measured within 350 m of the water bodies [2]. Among landscape types, water zones reduced the daily average air temperatures by 3.04 °C and yielded the maximum values of cooling effect (6.88), intensity (0.02), and gradient (0.99) in the Yangtze River Delta [17]. Surrounding urban morphology such as building density, vegetation cover and spatial proximity to water and modulates these effects [18]. In addition, it is also found that

simulations with the General Lake model indicate that floating PV (FPV) installations may lower thermal stratification, which could enhance water body's ability to adapt to changes in climate scenarios [5].

Landscape indices are the important tools of spatial pattern and ecological implications. ESVs of landscape quality in the Yihe River Basin [18] were negatively correlated with PD and ED. Spatial variability of the influence of diversity metrics such as Shannon's Diversity Index (SHDI) and Shannon's Evenness Index (SHEI) on ESVs was found, in [18, 19]. Rather, anthropogenic land modification was found to decrease vegetation complexity in Hungary, against the assumption of diversity benefits [20]. The implication of these findings is that the climatic parameters should be integrated into the landscape evaluation frameworks for improved accuracy and desirability in ecosystem-based planning [21]. The key landscape indices along with their definitions, formulas, input parameters used for computation are listed in Table 1.

Effect of Landscape on Land Surface Temperature (LST)

Land Surface temperature (not to be confused with warmth of the land) is a function of climate and landscape characteristics that have a strong impact on urban micro climates and reduction of Urban Heat Island (UHI) effects. For example, urban water bodies exhibit strong cooling effect, decreasing LST from place to as much as 350 m. Bubbling the green spaces together with these effects increases urban climate resilience and livability [2]. Additionally, seasonality is also found in the relationship among climatic variables and landscape factors. Elevation, slope, humidity, wind speed and vegetation cover are natural geographical attributes, which greatly influence LST in the transitional seasons, while socio economic factors play important roles in the summer season and landscape pattern has great influence on LST in the winter season [26].

It is important that urban greenspaces provide the shading, evapotranspiration and reduced thermal storage to reduce LST. Particularly, forests have higher cooling efficiency compared to any other land use [18]. Nevertheless, climatic and geographic contexts serve to affect their performance. Vegetated LCZs in subtropical cities reduced LST by 1.92°C over three years [27], but they have had less apparent effect on LST so far in arid cities such as Riyadh [28]. Broadleaf trees have greater cooling efficiency than grass or green roofs, reducing local air temperatures an average of 2 – 3 times more than grass or green roofs [29].

In addition, vegetation indices like evapotranspiration, density and shade outperform landscape metrics in UHI mitigation [22]. Complex fragmented green patches with high edge and patch density have stronger cooling potential than larger aggregated spaces [30]. For instance, thermal regulation capacity by greenspaces depends on spatial configuration [31, 32, 33] as well as area [34, 35]. Designing of adaptive, climate sensitive urban landscape [36] requires such understanding of these dynamics.

Recent research in the Indo-Gangetic plains has added useful knowledge regarding hydrogeomorphology, precipitation patterns, fire behavior, and wetland mapping [38-43]. Such studies nevertheless focus principally on spatial and temporal trends within single climatic zones and hardly incorporate climatic variability as an explanatory factor.

Conversely, this research presents a quantitative scaling model based on the radius of saturation (R_{Sat}) for standardizing water body areas and assessing their thermal impact under varying climatic conditions. By establishing a

connection between LST-derived cooling zones and climate sensitivity, we transition beyond descriptive mapping to formulate an applicable design principle for urban planners and practitioners of climate adaptation. Moreover, the use of

IDW interpolation with transect-based thermal analysis enables a finer-grained spatial resolution compared to conventional zone-based averaging.

Table 1. Landscape indices and parameters used to derive them

Index	Formula	Parameters	Reference
Landscape Shape Index (LSI)	$LSI = P / (2 * \sqrt{\pi * A})$	Perimeter (P), Area (A)	[22]
Shannon's Evenness Index (SHEI)	$SHEI = H' / H'max$	Shannon Diversity Index (H'), Maximum Shannon Diversity Index (H'max)	[23]
Shannon's Diversity Index (SHDI)	$SHDI = -\sum(\pi * \ln(\pi))$	Number of patch types (S), Proportion of landscape occupied by patch type (π)	[24]
Configurational Edge Density (ED)	$ED = E / A$	Total edge length (E), Landscape area (A)	[21]
Patch Density (PD)	$PD = (N / A) * 10,000$	Number of patches (N), Landscape area (A)	[25]

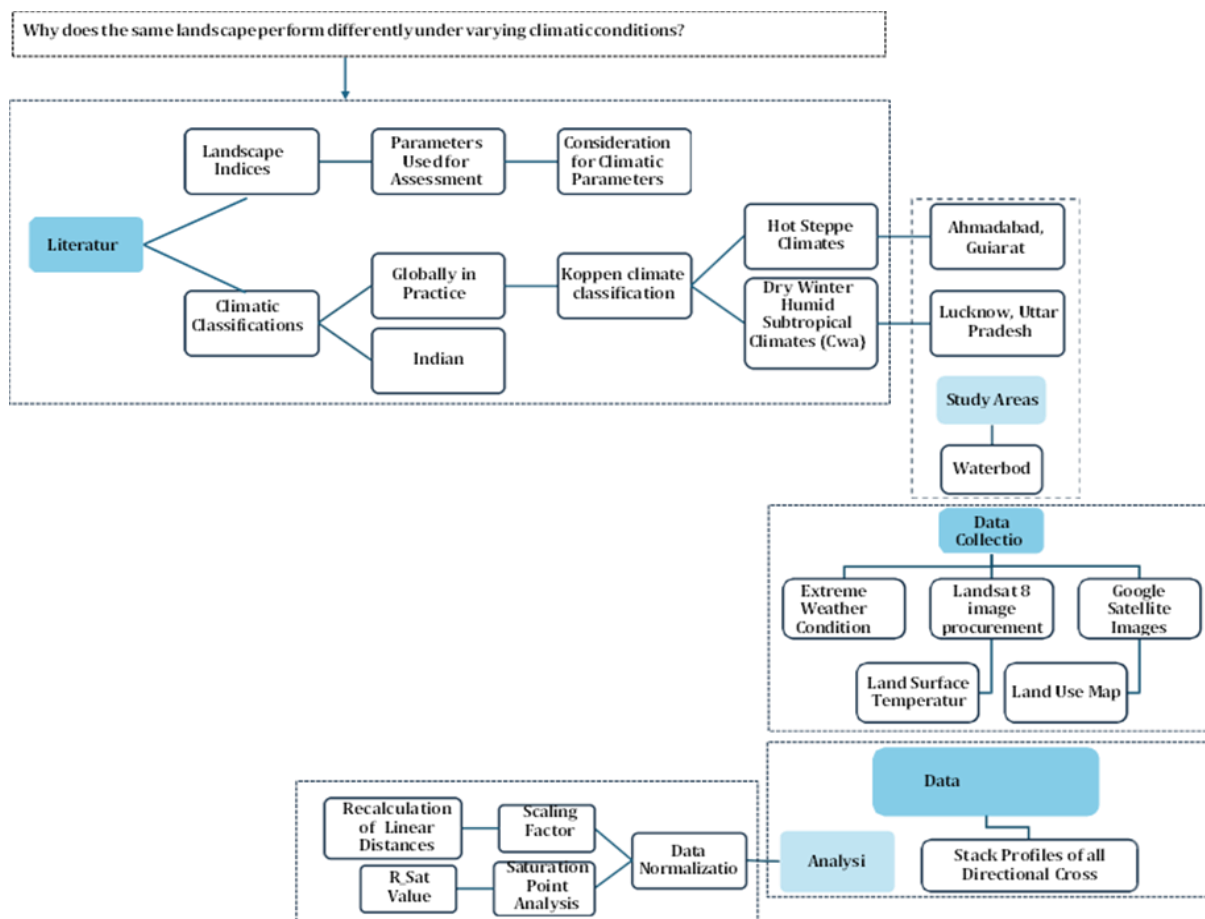


Figure 1. Detailed methodology

MATERIALS AND METHODS

In this research, the problem is related to the why performance behavior of the same landscape is different under different weather conditions. First, literature is reviewed thoroughly for all parameters of landscape indices, assessment parameters, and climatic considerations as well as the Koppen climate classification, after which the study begins. Two study areas namely Hot Steppe Climate (Bsh) of

Ahmedabad and Dry Winter Humid Subtropical Climate (Cwa) of Lucknow are chosen. Data were collected by looking at extreme weather conditions, Landsat 8 images for land surface temperature, Google satellite images for land use. Stack profiles of all directional cross section were created and the data collected processed since this formed the analysis. Normalization of the data was done, scaling factors were determined, saturation point analysis was performed, R_Sat values were calculated and linear distances were re-

calculated. The steps were to standardize the thermal influence of bodies of water in different climatic zones for the purpose of comparative analysis and of recommending the amendment of their cooling effects in differing climatic conditions. While the detailed Methodology can be found in Figure 1.

This figure presents a comprehensive workflow of the methodology adopted in the study. It outlines the various stages of data collection, preprocessing, and analysis, including remote sensing techniques, GIS-based spatial evaluations, and in-situ hydrological measurements. The methodology incorporates temperature and land-use datasets, computational models for linear distance calculations, and analytical techniques for water body profiling. This approach is structured so that the hydrological and climatic characteristics of the study regions are systematically assessed.

Study Area

Research methodology is the most important part in the research pertaining to select the study area, wherein we investigate the urban water bodies in two climatic classifications in India. As to need to sample a definite climatic and geographic conditions common to the Indian subcontinent, it is the choice to identify specific cities and micro study sites. The selection of Lucknow and Ahmedabad was done [37]. Köppen's climate system contains the classification of the study area into specific climatic zones that represented by figure 2. Such classification is very important for understanding the temperature variations, precipitation pattern and environmental influence on the water bodies in Lucknow and Ahmedabad.

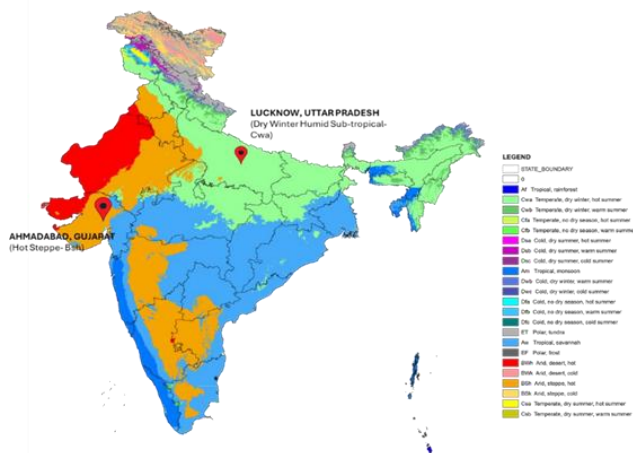


Figure 2. Demarcation of study area as per Köppen's climatic classification

For the study of the urban water body, this study based on an urban water body of Lucknow in Dry Winter Humid Subtropical (Cwa) and Ahmedabad in the Hot Steppe (Bsh) zone of Köppen climatic classification provides alternate and contrasting untry. In these cities, further Water bodies like lakes, ponds, and reservoirs were identified due to its important role in urban micro climates. These water bodies serve as heat sinks and cool the urban heat island by evaporative cooling effects. Specifically, two water bodies in Lucknow, Buttler Lake and Katotha Zeel, and one in Ahmedabad, Kankaria. Figure 3 shows the geographical location of Lucknow City in Uttar Pradesh for spatial context

of the study area. Figure 4 shows the demarcation of Ahmedabad City in Gujarat to locate Kankaria Lake within the urban landscape. A map as shown in Figure 5 highlighting the precise locations of Katotha Zeel and Buttler Zeel within Lucknow, including their spatial relation to urban settlements, road networks, and green cover. A spatial representation of Kankaria Lake's location within Ahmedabad, showing surrounding infrastructure, nearby urban zones, and ecological significance. This Figure 6 is crucial for understanding the lake's impact on urban hydrology and temperature distribution.

Detailed analysis was done for Lake, and were selected for that. Significance in affecting local microclimates as well as their potential to contribute to understanding water body effectiveness in different climatic settings are the criteria used to select these sites. It guarantees that the entirety of how urban water bodies support cooling and heat mitigation is covered, in several diverse climatic zones. Table 2 presents the area, perimeter, and shape index (SHI) for the selected water bodies in Lucknow and Ahmedabad.

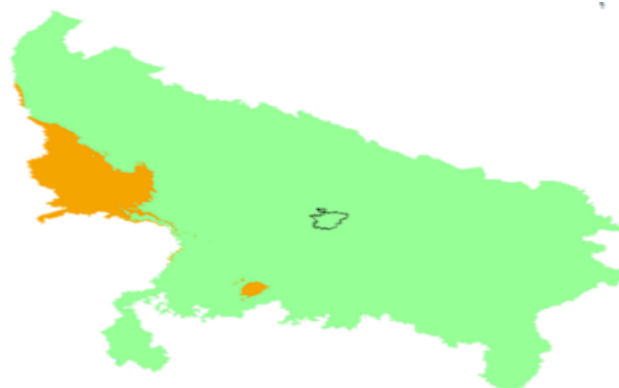


Figure 3. Demarcation of Lucknow City in State of Uttar Pradesh



Figure 4. Demarcation of Ahmedabad City in State of Gujarat



Figure 5. Location of Katotha Zeel & Buttler Zeel in Lucknow City

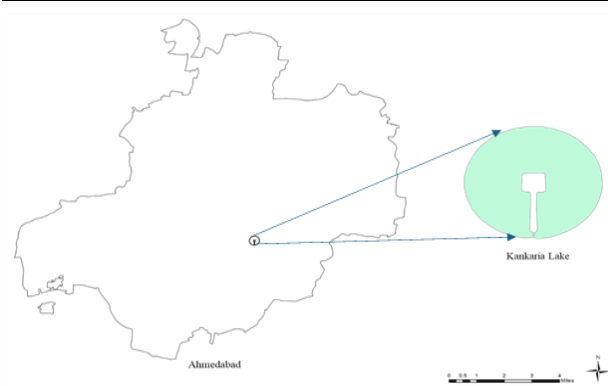


Figure 6. Location of Kankaria Lake in Ahmedabad City

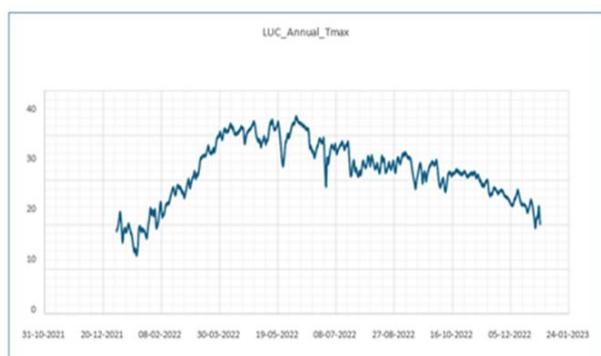


Figure 7. T max LUC 2022

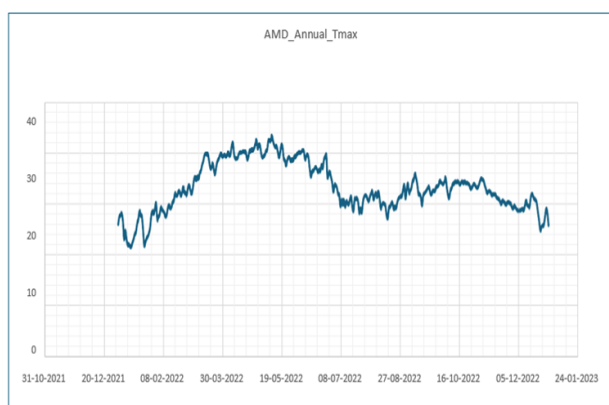


Figure 8. T max AMD 2022

Data Collection & Preparation

Extreme weather condition analysis

The collection of data started with the identification of the period when the peak summer was to be analysed for extreme weather conditions. This was done by analysing the data of annual maximum temperature (T max) that was obtained from the Indian Meteorological Department (IMD) libraries for Lucknow and Ahmedabad. Depending on the city, the periods of extreme summer conditions were identified through examination of historical temperature records. The importance of conducting this analysis is to make sure the data collection and analysis following would be on the extreme hot places. Ministry of Earth Sciences (MoES) focuses on the highest temperature phases to understand the urban heat island effect in its entirety. From March 31, 2022 to June 15, 2022, the peak summer day is June 4, 2022, and hence the SPP for Lucknow is determined to be a span from March 31,

2022, to June 15, 2022, as is shown in Figure 7. The SPP extension period for Ahmedabad is March 12, 2022 onward till June 30, 2022 (Figure 8), and peak summer day is May 11, 2022 (Figure 8) (Pune, 2022).

Satellite image procurement

After the summer peak periods (SPP) were identified, satellite imagery was obtained for the detailed analysis of land surface temperature (LST). From the United States Geological Survey (USGS) website, Landsat 8 images were downloaded while additional high-resolution image for the observed peak times were culled from Google Earth Pro. These images are used as a basis of analysis for the spatial distribution of land surface temperatures as well as how urban water bodies influence them (USGS, 2024).

For Lucknow it was dated April 30, 2022 and for Ahmedabad may 11, 2022 i.e. their peak summer days. High resolution imagery from Google Earth Pro too was acquired to complement the Landsat data, especially to get a more detailed view of the urban landscape and nearby landscape. The SPP dates and Landsat image procurement timeline for both cities are summarized in Table 3.

Hydrographic network integration and sub-basin overlay

In order to enhance the spatial analysis accuracy, a detailed hydrographic network was created using high-resolution drainage layers obtained from the Shuttle Radar Topography Mission (SRTM) DEM and USGS Hydrological Dataset. The hydrography was treated with ArcGIS Hydrology Tools, Flow Direction, Flow Accumulation, and Stream Definition functions to identify main and secondary drainage lines.

The completed network was subsequently overlaid on demarcated sub-basin limits, which were derived through watershed delineation according to the same DEM to make it topographically consistent. This overlay allowed for proper visualization of how the water bodies and their thermal effect zones relate to natural drainage pathways so that more understandable understanding of potential hydrological interactions and flow connectivity can be obtained. This upgrade promotes spatial realism beyond basic map overlays by ensuring that cooling zones correspond exactly with physical hydrographic features.

Land surface temperature (LST) analysis

Using the acquired Landsat images, the analysis of LST was carried out with the ArcGIS software. Spatial analysis of the temperature changes of the study regions involved the processing of the satellite data to derive LST values.

Remote sensing data specifics

Thermal assessment of this study was conducted with Landsat 8 Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) data collected during peak summer periods, when land surface temperature (LST) measurements typically peak in the study areas throughout the year.

For Lucknow, the imagery was received on 21 May 2022, and for Ahmedabad, the imagery was received on 25 May 2022. Both receptions were obtained from USGS Earth Explorer, with reception times at local overpass around 10:47 a.m. (IST) for Lucknow (Path 144, Row 042) and 10:41 a.m. (IST) for Ahmedabad (Path 149, Row 044). The cloud cover for both images was less than 2%, with clear-sky thermal observations.

Table 2. Area statement for selected sites

City	Land Use Category	Site	Area (Sq.M.)	Perimeter (M)	SHI
Lucknow (LUC)	Water Body	Katotha Jheel	475077	5680	2.06
		Buttler Jheel	22960	836	1.38
Ahmedabad (AMD)	Water Body	Kankaria Lake	259748	2711	1.32

Table 3. SPP & Landsat procured time frame

City	Summer Peak Time	SPP	Landsat Procured
Lucknow	04-06-2022	31/03/2022 to 15/06/2022	30-04-222
Ahmedabad	11-05-2022	12/03/2022 to 30/06/2022	11-05-2022

The LST retrieval was performed using the Single-Channel algorithm, which is widely used for Landsat 8 TIRS Band 10 due to its robust performance in urban environments. The algorithm involves three primary steps:

1. **Conversion of DN to Top-of-Atmosphere (TOA) Radiance** using radiometric rescaling coefficients provided in the metadata (MTL file),
2. **Conversion of TOA Radiance to Brightness Temperature (BT)** using sensor calibration constants, and
3. **Correction for atmospheric emissivity** to derive actual surface temperature.

$$L_{\lambda} = M_L \times Q_{cal} + A_L$$

$$T_B = \frac{K_2}{\ln\left(\frac{K_1}{L_{\lambda}} + 1\right)}$$

$$LST = \frac{T_B}{1 + \left(\frac{\lambda T_B}{\rho}\right) \ln \varepsilon}$$

where:

- M_L and A_L are radiometric rescaling factors,
- Q_{cal} is the quantized calibrated pixel value,
- K_1 and K_2 are calibration constants for Landsat 8 TIRS Band 10,
- λ is the wavelength of emitted radiance,
- $\rho = h \times c / \sigma$ (with h = Planck's constant, c = speed of light, σ = Boltzmann constant), and
- ε is the land surface emissivity derived from NDVI-based classification.

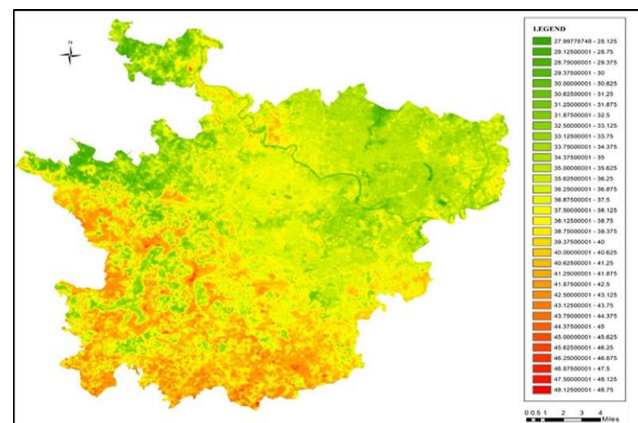
This method provides surface temperature retrieval with high reliability for urban microclimate studies. All processing was carried out in ArcGIS Pro 3.0 and QGIS 3.28, with additional atmospheric correction implemented through NDVI thresholding for emissivity estimation.

Integration of GIS Interpolation Techniques

To generate continuous surface and spatial temperature patterns in the focus urban water bodies, Inverse Distance Weighting (IDW) interpolation was used within the ArcGIS software. IDW was chosen because it is computationally efficient, deterministic, and well-suited to small to medium-

scale urban thermal mapping where any observation point has decreasing influence with distance. This approach is effective in targeting localized cooling effects and is not subject to the sophisticated statistical assumptions of Kriging. IDW is less sensitive to outliers compared to Spline and is thus more suitable for heterogeneous urban environments. Interpolated temperature surfaces generated were then subjected to analysis to map cooling influence areas and make thermal gradient estimates at each location.

The original satellite data was converted into useful ice temperature maps with the help of ArcGIS software since the thermal hotspots and interested locations surrounding the chosen water bodies were emphasized. The Land Surface Temperature (LST) map of Lucknow City (LUC) shows the differences in surface heat strength in various land-use categories. Figure 9 highlights the influence of the natural and built environments on temperature patterns, especially around water bodies and parks. An LST map of Ahmedabad showing spatial variations in temperature across the city. Figure 10 aids in appreciating the cooling effect of Kankaria Lake and surrounding greenery, making the study a foundation for sustainable urban planning.

**Figure 9.** LST Map of LUC

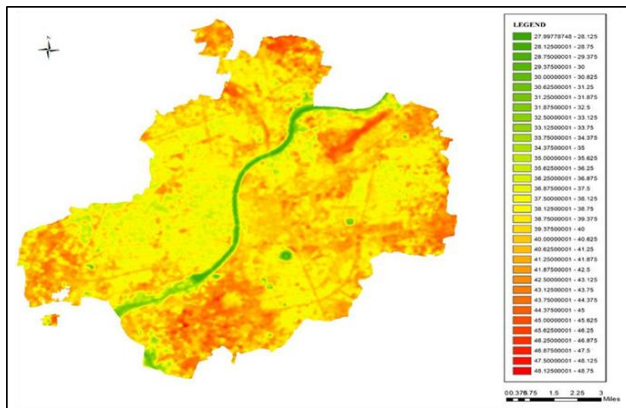


Figure 10. LST Map of AMD

Data sufficiency & robustness checks

This research measures thermal performance with deterministic surfaces (IDW in ArcGIS) and a scaling framework (R_Sat) instead of training a heavy-parameter predictive ML model. Although there are only a limited number of study locations (two in Lucknow; one in Ahmedabad), all locations offer high-granularity spatial data: (a) LST rasters (tens of thousands of pixels per scene) for peak summer seasons, (b) eight directional 45° cross-sections to measure distance-decay of cooling, and (c) buffer-based sampling around the boundaries of water bodies to estimate influence zones. To check robustness, we performed:

- Spatial cross-validation on transects: hold-one-direction-out tests to confirm stability of influence distances and ΔT profiles.
- Interpolation sensitivity analysis: varying IDW power ($p = 1.0-3.0$) and search radius; results retained rank-order of performance between climates.
- Sampling learning checks: progressively increasing pixel samples within buffers showed early stabilization of mean ΔT and influence distance estimates. These procedures support the adequacy of the dataset for comparative (climate-to-climate) inference on cooling performance. The cross-section design and influence-zone mapping are described earlier.

Observed land surface temperatures (29.56–46.43°C) exhibited thermal gradients in surface characteristics in relation to land cover composition for Lucknow. Water bodies, such as lakes and rivers, exhibited the lowest temperatures, ranging from 29°C to 34°C. Areas with dense vegetation, including agricultural lands, forests, and small urban vegetation patches, maintained relatively cooler temperatures between 34°C and 36°C. Conversely, the highest temperatures were found in barren agricultural lands in the southwest, ranging from 41°C to 45°C. Built-up areas displayed moderate to high temperatures, varying from 36°C to 40°C based on their compactness and land use patterns.

Land surface temperatures in Ahmedabad varied from 27°C to 48°C. Maximum temperatures were seen in concrete-paved airstrips or runways in the northeast and thick built-up areas with tin roofs in the southwest, and also in patches of barren land under direct sunlight exposure. Minimum temperatures were observed in water bodies like the Sabarmati River, and other large water bodies. Temperature in the high-density vegetation zones was lower, between 30°C and 33°C. In the urban environment, urban areas recorded temperature variations from 33°C to 43°C, based on land use and density of the building.

Individual site assessment

Land Use Mapping: Using the earlier purchased Google Earth Pro satellite images; the surrounding land use of every one of the sampled sites was mapped carefully. The mapping session provided data on the types and distribution of land use around the water bodies, which is significant in understanding the interaction of different land cover types and temperature distribution.

The site analysis on an individual level for the case of Ahmedabad's Kankaria Lake is used as a representative, and the same analysis was done for the other chosen water bodies. The initial land use mapping identified that to the north and northeast of the site, there are recreational areas in the form of parks and gardens to which they are an access boundary. Urbanized regions occupy the southern and eastern boundaries; regions of scattered patches of sparse vegetation patches with some patches of bare land are found in scattered locations. The classified distribution of land-use and land cover (LULC) in the vicinity of Kankaria Lake is presented in Figure 11. In this figure, water bodies, urban buildings, vegetation, and open land, are separated properly, which is crucial in determining anthropogenic impacts on lake hydrology.

Cross-Section and Buffer Zone Marking: Every 45-degree interval, the water body was projected outward and cross section were sketched. These sections covered eight main directions: east, west, north, south, northeast, southeast, northwest, and southwest. Influence radius of each location was also employed to mark buffer zones. It is the spatial scope of cooling influences provided by the water bodies that this step is essential to map. A topographic view of cross-sections chosen for Kankaria Lake, representing depth and width variation across different transects. This Figure 12 is used as a base for further hydrological and geomorphological assessment.

The minimum temperature in these cross sections was 28.9°C at the water body itself. In contrast, the adjacent railway yard and bare land had maximum temperatures of 43°C to 45°C. Vegetated areas had temperatures in the range 35°C to 37°C and 38°C to 40°C for built-up areas. This difference shows the large cooling by the water body in comparison with other land surfaces, i.e., land covered with heavy human construct and no vegetation.

The stack profile data were exported to Excel in order to do further analysis and graphical representation of temperature data. Temperature trends and pattern can be more easily identified with the help of graphical visualization and the point of occurrence can be identified of spatial hotspots and thermal gradients. This step converts cumbersome temperature data into graphs which are easy to understand and aids the data driven conclusions. The stack profile presentation of Kankaria Lake shows elevation variations and depth variations along chosen cross-sections in figure 13. This is important to understand the morphometric properties of the lake and its possible storage capacity of water.

The Table 4 data lends itself to identifying the role of water bodies in shaping the local microclimates at peak summer periods (SPP). An example of a cooling effect of SPP is the one in Ahmedabad, Kankaria Lake in the Hot Steppe (BSh) climatic zone which has a SPP of 29.1°C. Both Katotha Zeel and Buttler Zeel based in Dry Winter Humid (Cwa) and Subtropical (Cwa) climatic zone respectively, have SPP value of 29.58°C and 32.75°C respectively in Lucknow.

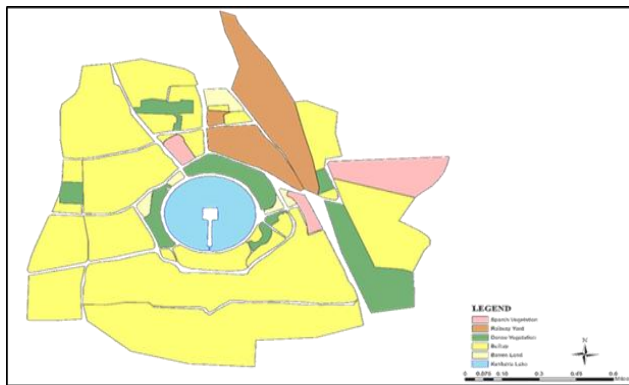


Figure 11. Land-use & land cover map of Kankaria Lake (WB AMD)

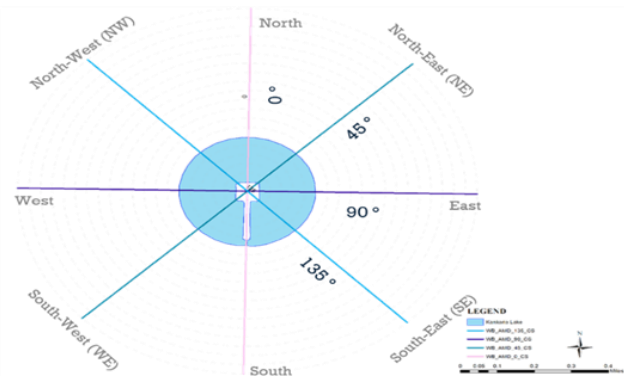


Figure 12. Demarcation of cross-sections of Kankaria Lake (WB AMD)

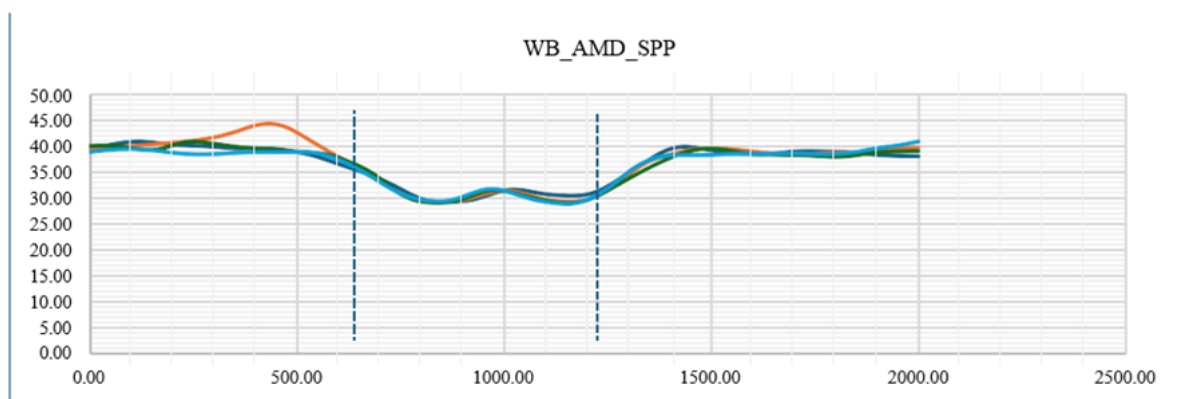


Figure 13. Generated Stack Profiles of Kankaria Lake (WB AMD)

Table 4. Water bodies & there SPP temperatures

City	Climatic Zone	Site	SPP (°C)
Ahmedabad	Hot Steppe (Bsh)	Kankaria Lake	29.1
Lucknow	Dry Winter Humid	Katotha Jheel	29.58
	Subtropical (Cwa)	Buttler Jheel	32.75

Data Analysis

Data normalization for comparative temperature Analysis by applying scaling (linear)

A scaling method was applied to normalize linear distances of water bodies from these climates to normalize size bias in the analysis of their impact on land surfaces temperature (LST). The normalization process for this normalization process is to see to it that the thermal impact on each individual water body is subjected to the same kind of influence in order to be able to draw any comparison in otherwise different climates.

Selection of cross-section

All eight principal directions were covered by the creation of cross sections at 45-degree intervals from North (0 degrees). A cross section was chosen based on its suitability concerning the surrounding urban environment where emphasis was put on building up areas concerning the control of urban thermal dynamics. Figure 14 shows a particular cross-section chosen for Katotha Zeel in Lucknow. Cross-sectional analysis assists in finding water depth changes, accumulation of sediments, as well as bank stability. These findings are critical to

hydrological evaluation and planning for conservation in the future for the water body. Figure 15 clearly shows a cross-section of Buttler Zeel. The profile reflects depth distribution and variations in slope, giving information on the morphology of the lake. An understanding of these variations becomes essential in water body management as well as counteracting the impacts of urban encroachment. This Figure 16 presents a chosen cross-section of Kankaria Lake, where changes in depth and topographic features are highlighted. The visualization assists in measuring sedimentation trends and hydrological behavior, providing important inputs for lake conservation measures.

Thus, multiple cross sections have been chosen for each water body, for a more accurate analysis that accounts for the influence of different surrounding land use. For Buttler Zeel, the 90-degree East cross section was chosen in Lucknow, whereas for Katotha Zeel, 45-degree West cross section was chosen. These directions depict the major developed area circumscribing the sites. Similarly, for Kankaria Lake in Ahmedabad, 0-degree South and 135-degree Southeast cross sections were chosen. The densely built-up urban areas around the lake are described by these cross sections and the lake's cooling effects on Built up may be evaluated in detail.

Table 5 lists the selected directional cross-sections for each water body used for thermal impact analysis.



Figure 14. Selected C/S for Katotha Zeel (WB_1 LUC)

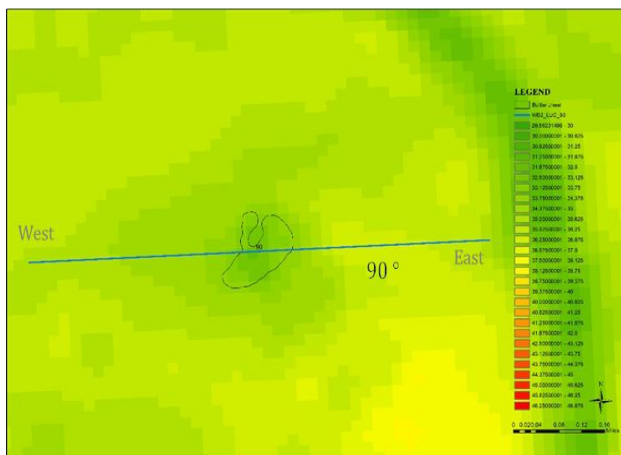


Figure 15. Selected C/S of Buttler Zeel (WB_2 LUC)

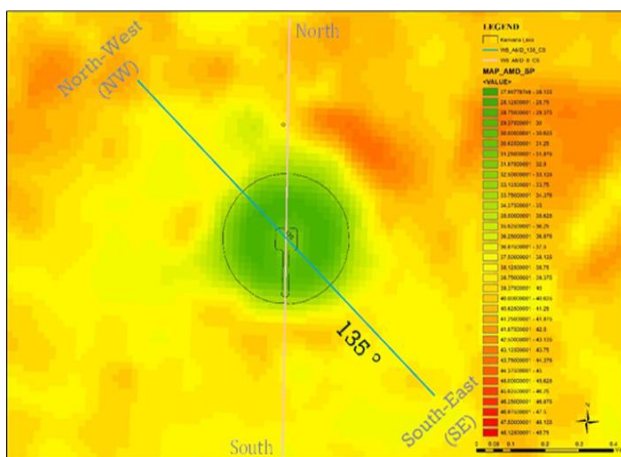


Figure 16. Selected C/S of Kankaria Lake (WB AMD)

Table 5. Sites & Selected C/S

City	Climatic Zone	Site	C/S Selected
Lucknow	Dry Winter Humid Subtropical (Cwa)	Kotatha Jheel	45 (SW)
		Buttler Jheel	90 (E)
Ahmedabad	Hot Steppe (Bsh)	Kankari a Lake	0 (S) 135 (SE)

Calculation of linear distances

The actual radius from the centre to the boundary and the influence distance at which the temperature case has an impact were calculated. The linear distance calculation for cross-section 45-SW of Katotha Zeel in Lucknow is presented in this Figure 17. The measurements illustrate horizontal and vertical spatial variations, which are crucial for understanding water distribution patterns and shoreline stability. A computed linear distance analysis for cross-section 90-E of Buttler Zeel in Lucknow is shown in this Figure 18. The calculations help in assessing water spread dynamics, bank stability, and the impact of external environmental factors on the water body.

Katotha Jheel:

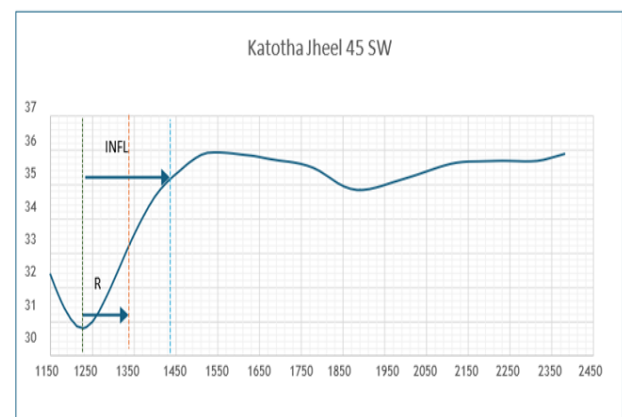


Figure 17. Calculation of linear dist. WB_1 LUC (C/S 45-SW)

Buttler Zeel:

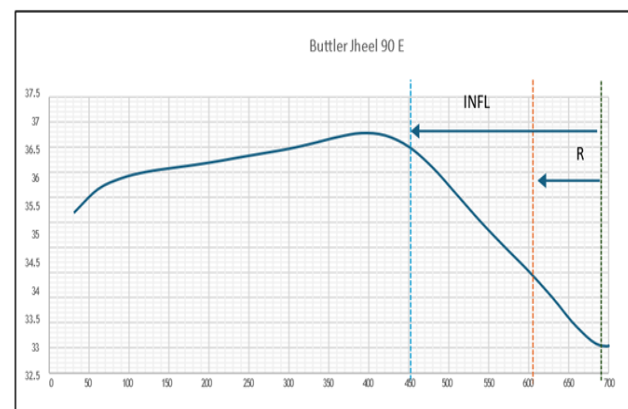


Figure 18. Calculation of linear dist. WB_2 LUC (C/S 90-E)

The linear distance computation for cross section 135-SE of Kankaria Lake in Ahmedabad is represented in this Figure 19. The analysis is useful for hydrological modelling and challenges associated with erosion patterns along the shore line. The insights provided here are good for sustaining water resource management. A detailed representation of the linear distance calculation for cross-section 0-S of Kankaria Lake is depicted in this Figure 20. The measurement provides insights into depth fluctuations and bank morphology, which are essential for lake restoration and flood mitigation planning.

For Buttler Zeel and Katotha Zeel both of Lucknow's teams, the average radius of 90 meters was found. These sites had average influence distances that averaged at 120 meters for the extent of temperature impact. In Ahmedabad, the analysis focused on Kankaria Lake, where the average radius was significantly larger at 300 meters. The influence distance for Kankaria Lake averaged at 137.5 meters. Average radii and influence distances of each water body are compiled in Table 6.

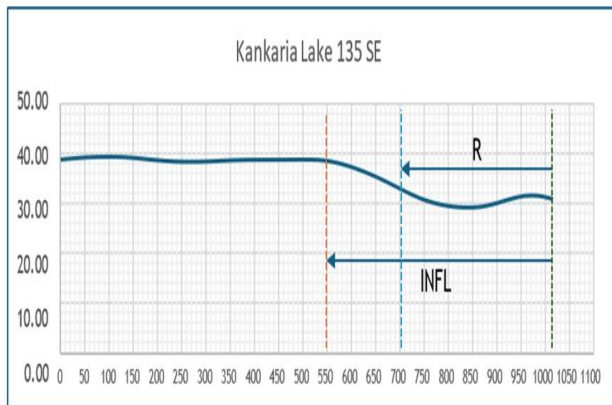


Figure 19. Calculation of linear dist. WB AMD (C/S 135-SE)

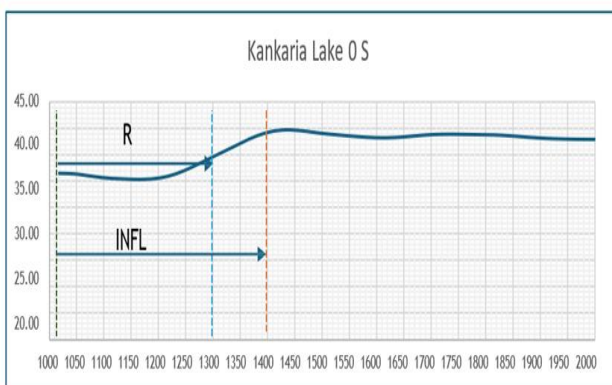


Figure 20. Calculation of linear dist. WB AMD (C/S 0-S)

Scaling

Determining Scaling Factor: A reference linear distance was used as a baseline for normalization.

Application of Scaling Factor: Scaling factors were used to scale the distances in cross-sectional temperature profiles. Every distance was scaled by the corresponding scaling factor

so that the physical dimensions of the water bodies were standardized.

Scaled Influence = Actual Influence × Scaling Factor

Calculation of Cross-Sectional Profiles: Cross-sectional temperature profiles were re-calculated with the scaled distances, representing normalized temperature effects as if all the bodies of water were the same linear distance from the terminals, allowing for direct comparison between climatic zones.

Saturation Point Analysis & Determining The R_{Sat}: Site temperature deviation with distance reduces, and follows an S-curve trend, with stabilization point being the distance beyond which temperature variation reduces to insignificance. Cross-sectional temperature profiles for every site were examined in order to determine the influence distance at which the temperature stabilizes, as the saturation point. Larger sites showed greater temperature changes to a plateau, the saturation point at which additional increases in site size have no influence on ΔT . Here, the radius is referred to as the radius of saturation. If the water body's size is larger than its radius of saturation, when scaling, the true radius needs to be replaced by its corresponding radius of saturation. i.e. If $R < R_{sat}$, we employ R . If $R > R_{sat}$, we employ R_{sat} . The saturation point defined the maximum scaling influence distance that allowed true thermal effects to be represented in a normalized comparison free from climatic prejudices. The recalculated distances in cross-sections and percentage-wise thermal performance of each site are presented in Table 7.

Mathematical definition of core metrics

To ensure methodological transparency and reproducibility, the calculation of the radius of saturation (R_{sat}), performance, efficiency, and scaling factor is presented below. These mathematical definitions form the basis of the comparative thermal analysis between Lucknow and Ahmedabad.

1. Radius of Saturation (R_{sat})

The radius of saturation represents the distance from the edge of the water body at which the cooling intensity becomes negligible. It is defined as:

$$R_{sat} = \min \{r \mid \Delta T(r) \leq \theta \times \Delta T_{max}\}$$

where:

- $\Delta T(r) = T_r - T_0$ is the temperature difference between the point at distance r from the edge of the water body and the temperature at the edge of the water body.
- ΔT_{max} is the maximum cooling intensity at the water body edge.
- θ is the threshold fraction at which the cooling effect is considered negligible.

In this study, $\theta = 0.1$ (i.e., 10%) was adopted based on the flattening of the temperature decay curve observed in the LST transect profiles. This represents the point beyond which additional distance contributes minimally to thermal mitigation.

Table 6. All AVG linear distances

City	Climatic Zone	Site	Radius of WB on that C/S	Average Radius	Influence	Average Influence
Lucknow	Dry Winter Humid Subtropical (Cwa)	Buttler Jheel	90	90	150	120
		Katotha Jheel	90		90	
Ahmedabad	Hot Steppe (Bsh)	Kankaria Lake	300	300	125	137.5
			300		150	

Table 7. Recalculated C/S and % wise performance

City	R	R_Sat	Standard Radius	Scaling Factor	Actual Influence	Scaled Influence	% wise performance
Lucknow	90	90	90	1	120	120	100
Ahmedabad	300	154	90	0.58	137.5	80.36	67

2. Thermal Performance

Thermal performance measures the relative strength of the cooling influence of a site compared to the benchmark site. It is defined as:

$$\text{Performance}(\%) = \frac{I_{\text{scaled}}}{I_{\text{benchmark}}} \times 100$$

where:

- I_{scaled} is the scaled influence distance of the water body after applying the R_{Sat} -based normalization.
- $I_{\text{benchmark}}$ is the influence distance of the highest performing site (Lucknow in this case).

3. Thermal Efficiency

Thermal efficiency measures the actual influence of a site relative to the benchmark. It is expressed as:

$$\text{Efficiency}(\%) = \frac{I_{\text{site}}}{I_{\text{benchmark}}} \times 100$$

where I_{site} is the measured influence distance of the site.

For Ahmedabad, the calculated efficiency is approximately 67% relative to Lucknow (100%).

4. Scaling Factor

The scaling factor indicates the required increase in the water body size for a lower-performing site to match the cooling performance of the benchmark site. It is given by:

$$\text{Scaling Factor} = \frac{I_{\text{benchmark}}}{I_{\text{site}}}$$

For this study:

$$\text{Scaling Factor} = \frac{120 \text{ m}}{80.36 \text{ m}} \approx 1.49$$

This means that water bodies in Ahmedabad would need to be increased by approximately 1.49× to achieve thermal performance equivalent to that of Lucknow.

5. Application of Metrics

With these mathematical equations, the saturation radius was calculated for every site, and then performance and efficiency calculations were made. The cutoff fraction $\theta=0.1$ was used for all transects uniformly to maintain

uniformity. The achieved performance values were 100% for Lucknow and 67% for Ahmedabad. The scaling factor of 1.49 is a useful planning parameter for climate-responsive water body design in heat-stressed cities.

Influence of Landscape Structure on Cooling Performance

Changes in the saturation radius R_{Sat} in Lucknow and Ahmedabad were highly correlated with changes in landscape pattern measures (Table 1). Lucknow showed greater R_{Sat} (120 m) and lesser landscape shape index (LSI) and Shannon's diversity index (SHDI), reflecting more dense water body morphology and higher landscape variation in the matrix surrounding it. Such structural arrangement permitted more evenly distributed zones of cooling.

Ahmedabad possessed a lower R_{Sat} (80.36 m), where the LSI is higher (more irregular) and SHDI lower, denoting a more broken and homogeneous built-up area. Such an arrangement is likely to impede lateral heat diffusion, minimizing the water body's effective cooling area.

This indicates that landscape shape performs a mediation role in adjusting the blue infrastructure's thermal impact, other than just meteorological conditions.

To enable meaningful comparison of thermal influence between water bodies located in different climatic zones, the radii of water bodies were standardized to a common baseline of 90 m prior to performance analysis. The selection of this baseline is not arbitrary but grounded in (i) site characteristics, (ii) literature benchmarks, and (iii) methodological consistency.

Empirical Site Features

The mean radius of the observed water bodies in Lucknow and Ahmedabad varied between 85 m and 95 m, using digitized perimeter information from high-resolution satellite data. Having a baseline of 90 m matches well with the mean observed size, avoiding bias from outlier values and maintaining realistic scale relationships.

Literature and Design Benchmarks

Various researches on urban blue infrastructure have noted small to medium water bodies (70–100 m radius) to be key thermal moderators in dense urban structure [38-40]. A 90 m reference point thus represents a common planning and ecological scale that suits both hydrological behavior and practical dimensions for urban water design.

Analytical Consistency

A uniform standardized baseline enables direct scaling and comparison across climatic regions free from confounding effects caused by differences in water body sizes. All calculations being normalized against a 90 m reference, the derived performance and scaling factors represent climate-sensitivity, not geometric variation. This provides for a purer interpretation of how climatic context affects thermal influence. Briefly, 90 m baseline was chosen as a representative and scientifically based scaling reference, vindicated by site information, pertinent literature, and cross-climate comparison analytical needs.

RESULTS AND DISCUSSION

The information gained from the comparative study of urban water bodies of different climatic regions of India indicates their thermo influence. According to the results, thermal influence is maximum in Lucknow in water bodies. The finding indicates that the dry conditions combined with a mild humidity in Lucknow help in better cooling effects of water bodies. They justify that Lucknow is a dry winter humid sub tropical (Cwa) climatic zone that might be having specific environment as water bodies can enhance the cooling effect to such an extent as to result in a considerable reduction in their surrounding land surface temperatures (LST).

However, the performance of the water bodies in Ahmedabad is comparatively lower, indicating that climatic conditions of this area have resisted major thermal influences. Located in a Hot Steppe (Bsh) climatic zone with arid parameters having high temperatures and low humidity, Ahmedabad may have water bodies that may not aid in introducing cooling. The analysis indicates an about 67% thermal performance of water bodies in Ahmedabad as compared to Lucknow. This emphasizes the climate effect on how the water bodies are effective in mitigating urban heat.

For similar thermal impact across the climatic zone of Ahmedabad, some required adjustments need to be done. This is to scale down the water bodies according to the size of the highest performing site in Lucknow to standardize thermal influence of water bodies in Ahmedabad. Observing the 67% performance of Lucknow at Ahmedabad, the water bodies can be scaled from 67 % to about 1.49 times to mimic the thermal influence in Lucknow. It consists in increasing the radius of pre-existing water bodies by adding more water, or it is creating new water bodies with an appropriate size. This would help enhance the cooling effect of water bodies in Ahmedabad and consequentially cool down the city more effectively. Table 8 quantifies the necessary increase in water body radius in Ahmedabad to match Lucknow's cooling effect.

Table 8. Required increase in linear distances

City	Climatic Zone	Scaling Factor	% Performance	Required increase in radius
Lucknow	Dry Winter Humid Subtropical (Cwa)	1	100	Baseline
Ahmedabad	Hot Steppe (Bsh)	0.58	67	1.49 x

Comparison with Previous Studies

The findings of this study are consistent with prior research demonstrating the cooling influence of urban water bodies on land surface temperature (LST). For example, studies in Pune and Hangzhou have reported cooling effects extending approximately 300–350 m from the water's edge, which aligns closely with our observed influence distances for Lucknow (120 m) and Ahmedabad (137.5 m) after scaling. Likewise, prior studies in Riyadh and Beijing have established that climatic conditions significantly affect cooling efficiency, with improved performance in humid climates over arid environments.

Yet, in contrast to these studies that generally aimed at single-city or single-climate analyses, our research presents a scaling method for climate sensitivity with the radius of saturation (R_{Sat}) for normalizing water body area to contrast performance throughout climatic regions. Cross-climate comparison is a new contribution that allows more realistic climateadaptive design approaches. In addition, the use of IDW-based interpolation and directional transect profiling offers more spatial resolution than is found in many prior research investigations based exclusively on averaged temperature buffers. By incorporating both thermal performance measures and climatic variability, this work builds on existing knowledge about urban blue infrastructure performance in various climatic zones, providing new findings for urban planning and heat mitigation policy.

Linking Landscape Metrics to Thermal Performance

Part of the observed differences in thermal performance between Lucknow and Ahmedabad are traceable to landscape configuration attributes. Low shape complexity (low LSI) and compact water bodies facilitate greater outward heat flux modulation across more uniform areas, while high SHDI diverse surrounding land covers increase gradient-driven dispersion of cooler air. On the other hand, irregular geometry and uniform surrounding land uses may constrain effective convective flow and spatial scope of thermal influence.

These results agree with existing studies demonstrating that landscape pattern regulates microclimate behavior and influences the intensity of cooling corridors within urban environments [38–43]. By integrating these indices into R_{Sat} interpretation, the study delivers an n-dimensional representation of how climate and landscape architecture influence cooling performance together.

Limitations and Future Data Expansion

We recognize that the pool of water bodies is finite. Nevertheless, the inference exploits dense spatial measurements (buffer sampling, multi-directional transects, and LST rasters) and deterministic mapping instead of ML training and thus has less reliance on huge sample numbers. Future research will diversify the site inventory over other Köppen zones, add multi-season LST composites, and compare different interpolators (e.g., Ordinary Kriging with variogram modeling) to better confirm climate-sensitive

scaling of cooling performance. Even though the present study offers a systematic approach to assess the climate-sensitive thermal behavior of urban water bodies, some sources of uncertainty need to be considered. To begin with, satellite-retrieved LST observations inherently contain radiometric and atmospheric correction errors, which can potentially affect the accuracy of surface temperature values, particularly in highly built-up urban areas. Second, IDW interpolation—albeit good for localized temperature gradients—fails to simulate spatial autocorrelation with the same stringency as geostatistical approaches such as Kriging, which might impact uncertainty estimates in transitional buffer areas. Third, the scaling model depends on consistent peak summer conditions and does not necessarily reproduce short-term climatic variability such as wind patterns, humidity anomalies, or heatwave extremes. Fourth, representativeness bias can be induced by the limited number of sites, although partly balanced by the high spatial density of data gathered within each site. Finally, the model does not currently quantify these uncertainties; future work will include uncertainty quantification, sensitivity analysis, and multi-seasonal validation in order to further develop confidence in the findings.

Practical Implications

The results give direct recommendations for urban landscape planning and climate adaptation in warm cities. This research measures the influence of climatic conditions on the thermal performance of water bodies and presents a practical decision-support system for city planners, landscape architects, and policymakers. To provide the same cooling effect as in humid subtropical climates such as Lucknow, water bodies in dry climates such as Ahmedabad need to be increased by 1.49×, based on scaling study. This is a climate-sensitive guide to blue infrastructure intervention sizing. Taking the above concepts into urban master plans can also reduce energy loads, enhance outdoor thermal comfort, and guide environmentally sustainable water management practices.

CONCLUSION

The present research provides evidence that the thermal efficiency of city water bodies is highly climate-dependent. We examined the cooling effect of water bodies in two different climatic regimes, Lucknow and Ahmedabad, through Landsat-extracted LST, directionally transected data, and the R_Sat-based scaling model. The 100% and 67% efficiencies in Lucknow and Ahmedabad, respectively, prove that weather factors exert a considerable effect on cooling efficiency. IDW interpolation with scaling facilitated very high-resolution thermal profiling and cross-site standardization. The integrative approach is a methodological advance over prior single-climate studies. The article delivers an evidence-based design basis for blue infrastructure in urban heat reduction strategies through direct linkages between thermal performance and climate sensitivity. The future study could be the coupling of sophisticated hydrological models and climate projections with adaptive water resource approaches. For long-term water sustainability for both areas, policy interventions must be enhanced while interdisciplinary collaborations pursued.

DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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